

Write 10 examples based on Tokenization and Embeddings.

1) I LOVE AI

Tokens: ["I", "love", "AI"]

Embeddings:  $\begin{bmatrix} [0.12, 0.45, 0.78] \\ [0.91, 0.23, 0.66] \\ [0.34, 0.67, 0.89] \end{bmatrix}$

2) Hello world

T: "Hello", "world"

E:  $\begin{bmatrix} [0.21, 0.54, 0.32] \\ [0.77, 0.12, 0.48] \end{bmatrix}$

3) Machine Learning

T = "Machine", "Learning"

E =  $\begin{bmatrix} [0.65, 0.32, 0.91] \\ [0.43, 0.87, 0.25] \end{bmatrix}$

4) "Good Morning"

T = "Good", "Morning"

E =  $\begin{bmatrix} [0.11, 0.72, 0.39] \\ [0.84, 0.29, 0.67] \end{bmatrix}$

5) "Data Science"

T: "Data", "Science"

E:  $\begin{bmatrix} [0.56, 0.18, 0.74] \\ [0.92, 0.44, 0.31] \end{bmatrix}$



6) Deep learning model  
 $\hat{T} = \text{"Deep", "learning", "model"}$

$$E = \begin{bmatrix} (0.67, 0.23, 0.81) \\ (0.43, 0.87, 0.25) \\ (0.19, 0.58, 0.93) \end{bmatrix}$$

7) "chat gpt is useful"

$\hat{T} = \text{"chat", "GPT", "is", "useful"}$

$$E = \begin{bmatrix} (0.88, 0.41, 0.66) \\ (0.25, 0.79, 0.93) \\ (0.22, 0.55, 0.14) \\ (0.73, 0.92, 0.37) \end{bmatrix}$$

8) Unbelievable.

$\hat{T} = \text{"Un", "believ", "able"}$

$$E = \begin{bmatrix} (0.12, 0.45, 0.78) \\ (0.67, 0.23, 0.91) \\ (0.34, 0.89, 0.58) \end{bmatrix}$$

9) My name is mouni priya

$\hat{T} = \text{"My", "name", "is", "mouni", "Priya"}$

$$E = \begin{bmatrix} (0.12, 0.45, 0.78) & (0.67, 0.23, 0.91) \\ (0.34, 0.89, 0.56) & (0.88, 0.41, 0.66) \\ & (0.25, 0.79, 0.93) \end{bmatrix}$$

10) Welcome to cohort class

$\hat{T} = \text{"Welcome", "to", "cohort", "class"}$

$$E = \begin{bmatrix} (0.72, 0.31, 0.85) & (0.14, 0.56, 0.29) \\ & (0.91, 0.42, 0.67) & (0.38, 0.79, 0.53) \end{bmatrix}$$